Zeshui Song, 2/3/26, HW2

1.34 A steel pipe of 300-mm outer diameter is fabricated from 6-mm-thick plate by welding along a helix that forms an angle of 25° with a plane perpendicular to the axis of the pipe. Knowing that the maximum allowable normal and shearing stresses in the directions respectively normal and tangential to the weld are $\sigma = 50$ MPa and $\tau = 30$ MPa, determine the magnitude P of the largest axial force that can be applied to the pipe.

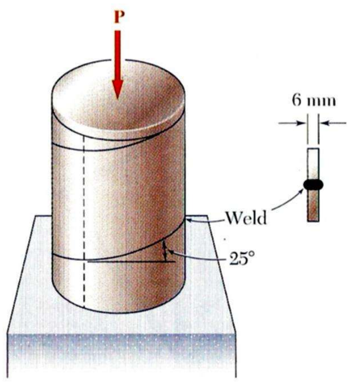


Fig. P1.33 and P1.34

Given:
- Outer diameter: 300 mm
- Thickness: 6mm
- Helix angle: 25 deg
- Max normal stress: 50 Mpa
- Max tangential stress: 30 MPa

Find:
- Magnitude of P when the weld fails

Approach:
- Using the shear and normal stress equations

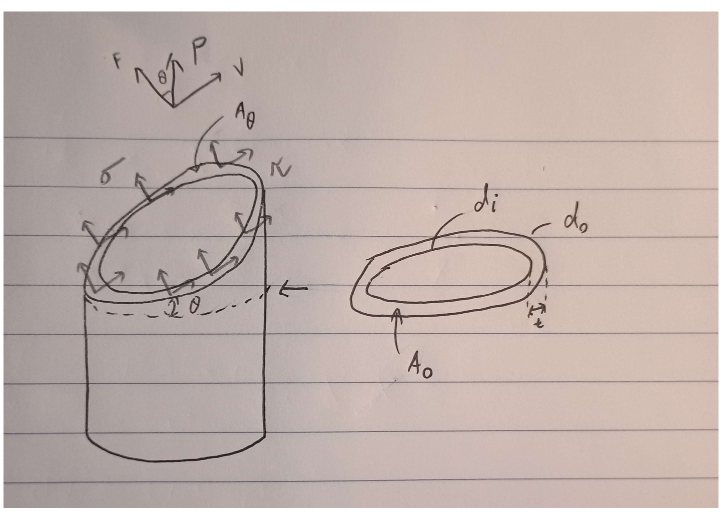
$$\sigma = \frac{P}{A_0}\cos^2\theta$$

$$\tau = \frac{P}{A_0} sin\theta cos\theta$$

I can solve for P given the max normal and tangential stress, picking the smaller max magnitude of P. The area A_0 will be the annulus perpendicular to the axial load.

1

Zeshui Song, 2/3/26, HW2



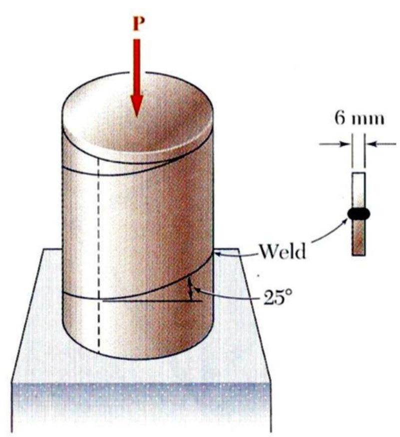


Fig. P1.33 and P1.34

2

Zeshui Song, 2/3/26, HW2

$$A_0 = \pi\left(\frac{d_o^2}{4} - \frac{d_i^2}{4}\right), d_i = d_o - 2t$$

$$A_0 = \frac{\pi}{4}(d_o^2 - (d_o - 2t)^2)$$

$$A_0 = \frac{\pi}{4}([300mm]^2 - ([300mm] - 2[6mm])^2) = 5541.76mm^2$$

Based on normal stress:

$$\sigma = \frac{P}{A_0}\cos^2\theta \rightarrow P = \frac{\sigma A_0}{\cos^2\theta}$$

$$P = \frac{[50\ MPa]\left[\frac{10^6 Pa}{MPa}\right][5541.76mm^2]\left[\frac{(10^{-3}m)^2}{mm^2}\right]}{\cos^2(25°)} = 337.33\ kN$$

Based on tangential stress:

$$\tau = \frac{P}{A_0} sin\theta cos\theta \rightarrow P = \frac{\tau A_0}{sin\theta cos\theta}$$

$$P = \frac{[30\ MPa]\left[\frac{10^6 Pa}{MPa}\right][5541.76mm^2]\left[\frac{(10^{-3}m)^2}{mm^2}\right]}{\sin(25°)\cos(25°)} = 434.05\ kN$$

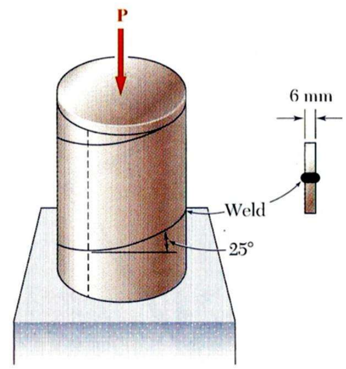


Fig. P1.33 and P1.34

$\boxed{\text{Picking the smaller load, } P = 337.33\ kN}$

This makes sense since the angle is steeper (closer to 0 deg), more of the stress will be normal, thus the axial load based on the max normal stress will be the determining factor.

Time spent: 40 mins

3

Zeshui Song, 2/3/26, HW2

1.49 A steel plate $\frac{5}{16}$ in. thick is embedded in a horizontal concrete slab and is used to anchor a high-strength vertical cable as shown. The diameter of the hole in the plate is $\frac{3}{4}$ in., the ultimate strength of the steel used is 36 ksi, and the ultimate bonding stress between plate and concrete is 300 psi. Knowing that a factor of safety of 3.60 is desired when $P = 2.5$ kips, determine (*a*) the required width a of the plate, (*b*) the minimum depth b to which a plate of that width should be embedded in the concrete slab. (Neglect the normal stresses between the concrete and the lower end of the plate.)

Given:
- Plate thickness: t = 5/16 in
- Hole diameter: d = 3/4 in
- Plate ultimate strength: σ_u = 36 ksi
- Ultimate bonding stress: τ_u = 300 psi
- Factor of safety: FS = 3.6
- P = 2.5 kips

Find:
- Required width of plate: a
- Minimum depth of plate in concrete: b

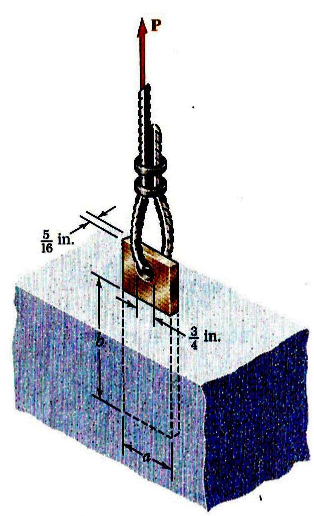
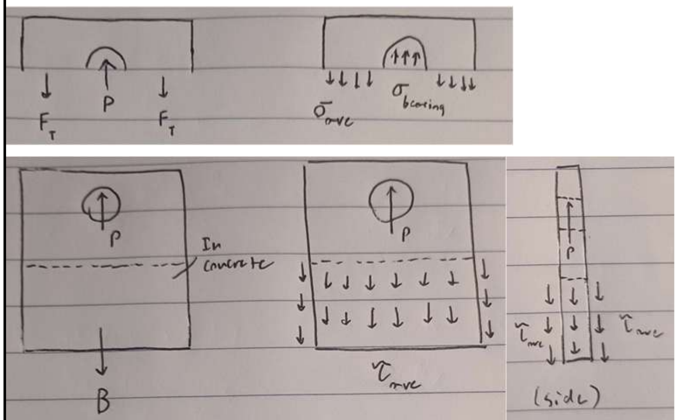


Fig. P1.49

4

Zeshui Song, 2/3/26, HW2

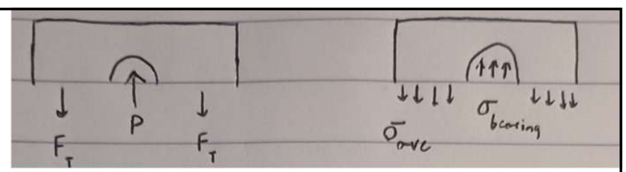


Approach:

- Using the factor of safety, I can solve for the allowable stress for the given P.
- Width of the plate can be found by solving for the cross-sectional area using the max average normal stress at the hole.
- Depth can be found by solving for the surface area under shear (full perimeter) in the concrete using the max allowable shear stress.

5

Zeshui Song, 2/3/26, HW2



$$FS = \frac{\sigma_u}{\sigma_a} \rightarrow \sigma_a = \frac{\sigma_u}{FS}$$

$$\sigma_a = \frac{\sigma_u}{FS} = \frac{P}{(a-d)t} \rightarrow \frac{\sigma_u t}{FS \cdot P} = \frac{1}{a-d} \longrightarrow a = \frac{FS \cdot P}{\sigma_u t} + d$$

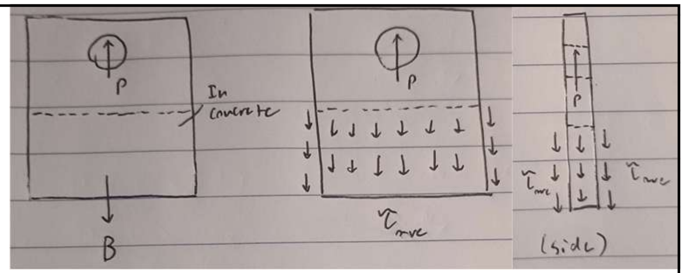
$$a = \frac{3.6\,[2.5\ kips]\left[\frac{1000\ lbf}{1\ kips}\right]}{[36\ ksi]\left[\frac{1000\ psi}{1\ ksi}\right]\left[\frac{1\frac{lbf}{in^2}}{1\ psi}\right]\left[\frac{5}{16}\ in\right]} + \left[\frac{3}{4}\ in\right] = 1.55\ in$$

$$\boxed{a = 1.55\ in}$$

6

Zeshui Song, 2/3/26, HW2

$$FS = \frac{\tau_u}{\tau_a} \rightarrow \tau_a = \frac{\tau_u}{FS}$$



$$\tau_a = \frac{\tau_u}{FS} = \frac{P}{2(ab) + 2(tb)} \rightarrow \frac{2\tau_u}{FS \cdot P} = \frac{1}{b(a+t)}$$

$$b = \frac{FS \cdot P}{2\tau_u\,(a+t)}$$

$$b = \frac{3.6\,[2.5\,kips]\left[\frac{1000\,lbf}{1\,kips}\right]}{2\,[300\,psi]\left[\frac{1\frac{lbf}{in^2}}{1\,psi}\right]\left[1.55 + \frac{5}{16}\,in\right]} = 8.05\,in$$

$$\boxed{b = 8.05\,in}$$

The width found was roughly twice as much as the diameter of the hole, which indicates a weaker spot. For the depth, including the sides as well as the front and back decreased the required depth. [Time spent: 40 mins]

7

Zeshui Song, 2/3/26, HW2 (GRADED PROBLEM)

1.51 Link AC is made of a steel with a 65-ksi ultimate normal stress and has a $\frac{1}{4} \times \frac{1}{2}$-in. uniform rectangular cross section. It is connected to a support at A and to member BCD at C by $\frac{3}{8}$-in.-diameter pins, while member BCD is connected to its support at B by a $\frac{5}{16}$-in.-diameter pin; all of the pins are made of a steel with a 25-ksi ultimate shearing stress and are in single shear. Knowing that a factor of safety of 3.25 is desired, determine the largest load **P** that may be applied at D. Note that link AC is not reinforced around the pin holes.

(Note that this problem assumes that BCD has been already designed to take a higher load with the desired factor of safety, so you do not need to consider stresses in BCD in this problem.)

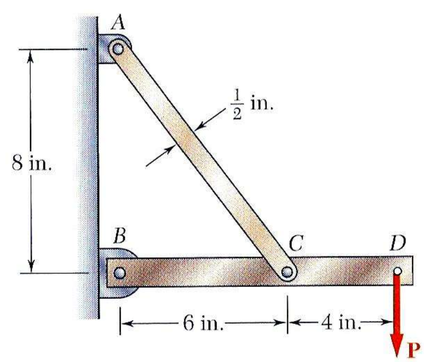


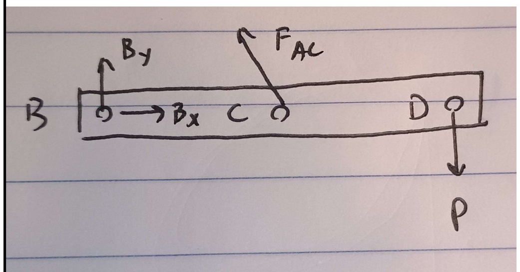
Fig. P1.51

Given:
- Link AC:
  - Ultimate stress: $\sigma_u = 65$ ksi
  - Width: $w = 1/2$ in
  - Thickness: $t = 1/4$ in
- Pin diameter (at A & C): $d_{AC} = 3/8$ in
- Pin diameter (at B): $d_B = 5/16$ in
- Pin ultimate shear stress (single shear): $\tau_u = 25$ ksi
- Factor of safety: $FS = 3.25$

Find:
- Largest load P that can be applied to D such that AC doesn't fail

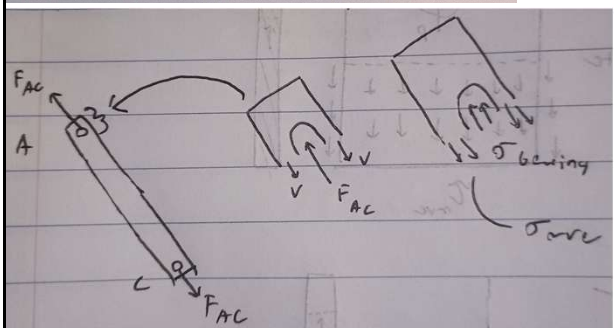
8

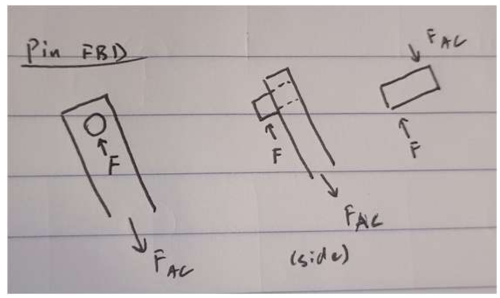
Zeshui Song, 2/3/26, HW2 (GRADED PROBLEM)



Approach:

- Using a moment balance at B, and Varignon's Thm, I can solve for the force on link AC in terms of P
- Using F_AC, I can then solve for the max avg normal stress and pin shear stresses (at A,C, and B) in terms of P. Comparing this to the max allowable stresses, I can solve for the max value of P.





9

Zeshui Song, 2/3/26, HW2 (GRADED PROBLEM)

$$\sum M_B = F_{AC,y}\,[6\ in] - P\,[10\ in] = 0\ ,\ F_{AC,y} = F_{AC}\frac{8}{\sqrt{8^2+6^2}}$$

$$\sum M_B = \frac{8}{10}F_{AC}\,[6\ in] - P\,[10\ in] = 0$$

$$\boxed{\frac{8}{10}F_{AC}\,[6\ in] = P\,[10\ in]\ \rightarrow\ F_{AC} = \frac{25}{12}P}$$

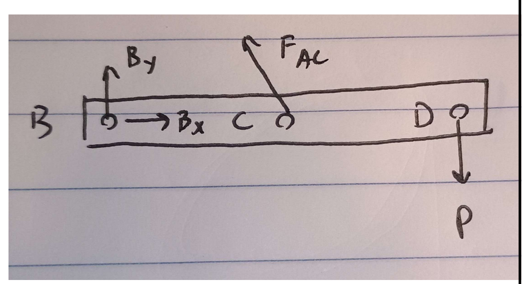
$$\sum F_x = -F_{AC,x} + B_x = 0\ ,\ F_{AC,x} = F_{AC}\frac{6}{\sqrt{8^2+6^2}}$$

$$\boxed{B_x = \frac{6}{10}F_{AC} = \frac{6}{10}\cdot\frac{25}{12}P = \frac{5}{4}P}$$

$$\sum F_y = B_y + F_{AC,y} - P = 0$$

$$\boxed{B_y = P - \frac{8}{10}F_{AC} = P - \frac{8}{10}\cdot\frac{25}{12}P = -\frac{2}{3}P}$$

$$\boxed{B = \sqrt{B_x^2 + B_y^2} = \frac{17}{12}P}$$



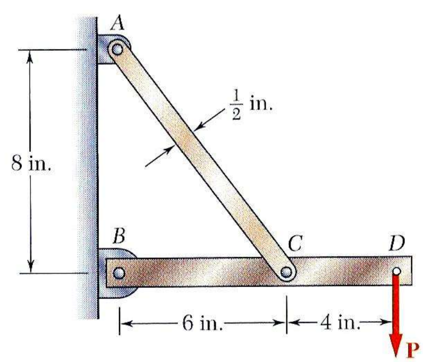


Fig. P1.51

10

Zeshui Song, 2/3/26, HW2 (GRADED PROBLEM)

Based on average normal stress in AC:

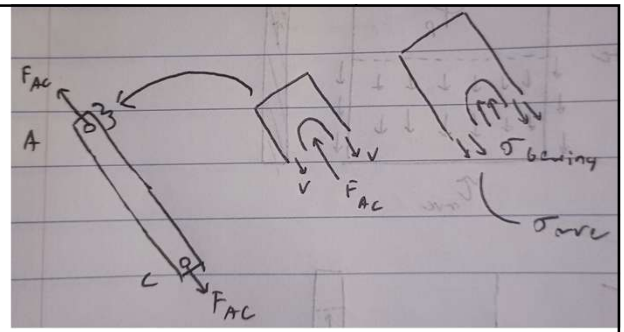
$$FS = \frac{\sigma_u}{\sigma_a} \rightarrow \sigma_a = \frac{\sigma_u}{FS}$$

$$\sigma_a = \frac{\sigma_u}{FS} = \frac{F_{AC}}{(w - d_{AC})t} = \frac{\frac{25}{12}P}{(w - d_{AC})t}$$

$$\downarrow$$

$$P = \frac{12}{25}\frac{\sigma_u}{FS}(w - d_{AC})t$$

$$P = \frac{12}{25}\frac{[65\ ksi]\left[\frac{1000\ psi}{1\ ksi}\right]\left[\frac{1\frac{lbf}{in^2}}{1\ psi}\right]}{3.25}\left[\frac{1}{2} - \frac{3}{8}\ in\right]\left[\frac{1}{4} in\right] = 300\ lbf$$



11

Zeshui Song, 2/3/26, HW2 (GRADED PROBLEM)

Based on single shear stress in pins at A and C:

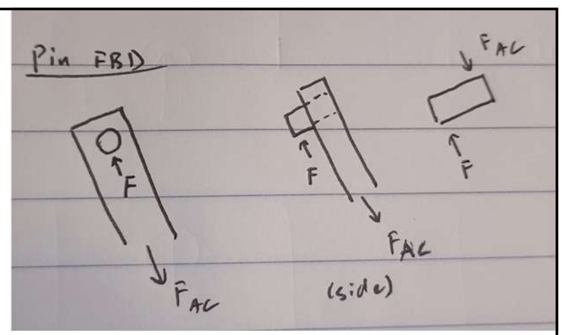
$$FS = \frac{\tau_u}{\tau_a} \rightarrow \tau_a = \frac{\tau_u}{FS}$$

$$\tau_a = \frac{\tau_u}{FS} = \frac{F_{AC}}{\frac{\pi}{4}d_{AC}^2} = \frac{\frac{25}{12}P}{\frac{\pi}{4}d_{AC}^2} = 4 \cdot \frac{25}{12}\frac{P}{\pi d_{AC}^2}$$

$$\downarrow$$

$$P = \frac{3\pi}{25}\frac{\tau_u d_{AC}^2}{FS}$$

$$P = \frac{3\pi}{25}\frac{[25\ ksi]\left[\frac{1000\ psi}{1\ ksi}\right]\left[\frac{1\frac{lbf}{in^2}}{1\ psi}\right] \cdot \left[\frac{3}{8}\ in\right]^2}{3.25} = 407.80\ lbf$$



12

Zeshui Song, 2/3/26, HW2 (GRADED PROBLEM)

Based on single shear stress in pin B:

$$FS = \frac{\tau_u}{\tau_a} \rightarrow \tau_a = \frac{\tau_u}{FS}$$

$$\tau_a = \frac{\tau_u}{FS} = \frac{B}{\frac{\pi}{4}d_B^2} = \frac{\frac{17}{12}P}{\frac{\pi}{4}d_B^2} = \frac{4}{\pi}\cdot\frac{17}{12}\frac{P}{d_B^2}$$

$$\downarrow$$

$$P = \frac{\pi}{4}\cdot\frac{12}{17}\frac{\tau_u d_B^2}{FS}$$

$$P = \frac{\pi}{4}\cdot\frac{12}{17}\frac{[25\ ksi]\left[\frac{1000\ psi}{1\ ksi}\right]\left[\frac{1\frac{lbf}{in^2}}{1\ psi}\right]\cdot\left[\frac{5}{16}in\right]^2}{3.25} = 416.46\ lbf$$

Picking the smallest load (from normal stress in AC), $P = 300\ lbf$

This makes sense since the hole diameter is a significant portion of the width of link AC (75%), which means that there was very little material around the hole, resulting in the weakest point. Additionally, while pin B is smaller than pins A and C, it sees a smaller load, which is why it can sustain a larger load. [Time spent: 45 mins]

13

Zeshui Song, 2/3/26, HW2

2.2 A 60-m-long steel wire is subjected to 6 kN tensile force. Knowing that $E = 200$ GPa and that the length of the rod increases by 48 mm, determine (*a*) the smallest diameter that may be selected for the wire, (*b*) the corresponding normal stress.

Given:
- Initial length: $l_0 = 60$ m
- Applied force: $F = 6$ kN
- Young's modulus: $E = 200$ GPa
- Change in length: $\delta l = 48$ mm

Find:
- Smallest diameter of the wire
- Normal stress

Approach:
- Using Young's modulus, I can find the normal stress. Since the applied force is given, then I can find the area and thus diameter.

$$E = \frac{\sigma}{\epsilon}, \epsilon = \frac{\delta l}{l_0} \rightarrow E = \frac{\sigma l_0}{\delta l} \rightarrow \sigma = \frac{E\delta l}{l_0} = \frac{F}{\frac{1}{4}\pi d^2}$$

$$\sigma = \frac{[200\ GPa]\left[\frac{10^9 Pa}{1\ GPa}\right][48\ mm]\left[\frac{10^{-3}\ m}{1\ mm}\right]}{[60\ m]} = 160\ 000\ 000\ Pa$$

$$\boxed{\sigma = 160\ MPa}$$

$$d = \sqrt{\frac{4F}{\pi\sigma}}$$

$$d = \sqrt{\frac{4[6\ kN]\left[\frac{10^3 N}{1\ kN}\right]}{\pi[160\times 10^6\ Pa]}} = 0.0069\ m$$

$$\boxed{d = 6.9\ mm}$$

Due to the large Young's modulus, it makes sense that such a large applied load was needed to stretch it just a little bit. [Time spent: 10 mins]

14

Zeshui Song, 2/3/26, HW2

2.4 Two gage marks are placed exactly 250 mm apart on a 12-mm-diameter aluminum rod with $E = 73$ GPa and an ultimate strength of 140 MPa. Knowing that the distance between the gage marks is 250.28 mm after a load is applied, determine (*a*) the stress in the rod, (*b*) the factor of safety.

Given:
- Initial length: l_0 =250 mm
- Diameter: d =12 mm
- Young's modulus: E = 73 Gpa
- Ultimate strength: σ_u = 140 MPa
- Final length: l_f =250.28 mm

Find:
- Stress in the rod
- Factor of safety

Approach:
- Using Young's modulus, I can find the normal stress. Since the ultimate strength is given, I can divide for the factor of safety

$$\delta l = l_f - l_0$$

$$E = \frac{\sigma}{\epsilon}, \epsilon = \frac{\delta l}{l_0} \rightarrow E = \frac{\sigma l_0}{\delta l} \rightarrow \sigma = \frac{E \delta l}{l_0}$$

$$\sigma = \frac{[73\ GPa]\left[\frac{10^9 Pa}{1\ GPa}\right][250.28 - 250\ mm]\left[\frac{10^{-3}\ m}{1\ mm}\right]}{[250\ mm]\left[\frac{10^{-3}\ m}{1\ mm}\right]} = 81760000\ Pa$$

$$\boxed{\sigma = 81.76\ MPa}$$

$$\boxed{FS = \frac{\sigma_u}{\sigma} \rightarrow FS = \frac{[140\ MPa]}{[81.76\ MPa]} = 1.712}$$

Since the part didn't fail after the load was applied, its reasonable to say that it is the allowable stress. [Time spent: 10 mins]

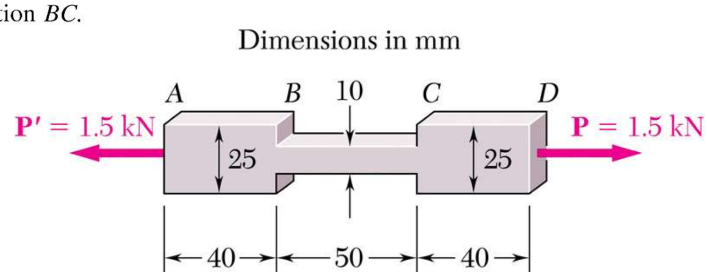
15

Zeshui Song, 2/3/26, HW2

2.15 The specimen shown has been cut from a 5-mm-thick sheet of vinyl ($E = 3.10$ GPa) and is subjected to a 1.5-kN tensile load. Determine (*a*) the total deformation of the specimen, (*b*) the deformation of its central portion *BC*.

Given:
- Thickness: t = 5 mm
- Young's modulus: E = 3.10 Gpa
- Applied force: F = 1.5 kN

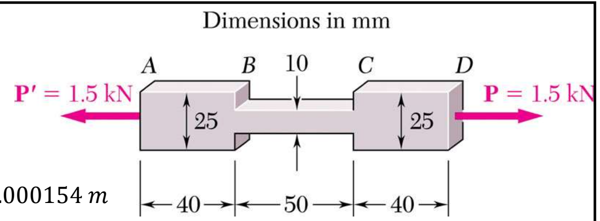
Find:
- Total deformation
- Deformation of BC



Approach:
- Using the deflection formula, I can solve for the deflection for section AB/CD and BC and sum it up for total deformation

16

Zeshui Song, 2/3/26, HW2



$$\delta_{AB} = \frac{PL_{AB}}{EA_{AB}} = \frac{PL_{AB}}{Ew_{AB}t}$$

$$\delta_{AB} = \frac{[1.5\ kN]\left[\frac{10^3 N}{1\ kN}\right][40\ mm]\left[\frac{10^{-3} m}{1mm}\right]}{[3.1\ GPa]\left[\frac{10^9 Pa}{1GPa}\right][25\ mm]\left[\frac{10^{-3} m}{1mm}\right][5\ mm]\left[\frac{10^{-3} m}{1mm}\right]} = 0.000154\ m$$

$$\delta_{BC} = \frac{PL_{BC}}{EA_{BC}} = \frac{PL_{BC}}{Ew_{BC}t}$$

$$\delta_{BC} = \frac{[1.5\ kN]\left[\frac{10^3 N}{1\ kN}\right][50\ mm]\left[\frac{10^{-3} m}{1mm}\right]}{[3.1\ GPa]\left[\frac{10^9 Pa}{1GPa}\right][10\ mm]\left[\frac{10^{-3} m}{1mm}\right][5\ mm]\left[\frac{10^{-3} m}{1mm}\right]} = 0.000483\ m$$

$$\boxed{\delta_{total} = 2\delta_{AB} + \delta_{BC} = 0.791\ mm}$$

$$\boxed{\delta_{BC} = 0.483\ mm}$$

Since segment BC is thinner, it makes sense that it deformed more than AB and CD. [Time spent: 10 mins]

17